AOU
الجامعة العربية المفتوحة
Arab Open University

Faculty of Business Studies

MIS200-IS

Relational Database Management Systems

Final Examination / Version: B

2025-2026 / Summer

1 – September – 2026

AOU EGYPT

Number of Exam Pages: 3 (including this cover sheet)	Time Allowed: 2 hours

Instructions:

1- Take **FIVE (5)** minutes to read all instructions carefully before attempting to write.
2- Mobile phones must be in "SWITCHED **OFF**" mode throughout the examination.
3- This is a **CLOSED** book examination. NO documents of any kind are allowed.
4- You **MUST** answer the questions using only the answer booklets provided
- You **MUST** return the examination paper together with your answer booklets.
- Do **NOT** write your answers in pencil except graphs/sketches
- Students are **NOT** allowed to use any type of dictionary or mobile phones during the examination
- Any form of cheating will **NOT** be tolerated and will subject you to AOU Cheating Policy.

الجامعة العربية المفتوحة - مصر
Arab Open University
القبول والتسجيل
الامتحانات
Arab Republic of Egypt

CamScanner

YOU HAVE TO ANSWER ALL THE QUESTIONS

Question 1 (35 marks)

Construct an E-R diagram for an online retail store that includes products and customers. Associate with each customer a history of their orders, including product details, order dates, and payment status.

Question 2 (35 marks)

Consider the following clinic appointment database, where the primary keys are underlined.

patient (patient_id, name, phone)
doctor (doctor_id, name, specialization)
appointment (appointment_id, patient_id, doctor_id, appointment_date)

a) **Write DDL statements for creating all tables with constraints. Make any reasonable assumptions about data types and be sure to declare primary and foreign keys (15 marks).**

b) **Write an SQL query to retrieve appointments with doctor name 'Dr. Ahmed' (5 marks).**

c) **Write an SQL query to insert a new patient record (assume the values) (5 marks).**

d) **Write an SQL query to delete an appointment using appointment_id (assume the appointment_id is known) (5 marks).**

e) **Write an SQL query to remove the entire doctor's table from the database (5 marks).**

YOU EGYPT

الجامعة العربية المفتوحة - مصر
Arab Open University
القبول والتسجيل
الامتحانات
Arab Republic of Egypt

CamScanner

Question 3 (30 marks)

a) **Explain how entities are transformed into tables (15 marks)**

b) **Choose the correct answer for each question below (15 marks):**

1) Given the table

CUSTOMER(CustID, Name, PhoneNum, AcctBalance)

What is the standard SQL query phrase to retrieve the Name and Phone Number of custome

A) SELECT Name, PhoneNum
B) SELECT (CUSTOMER-Name AND CUSTOMER-PhoneNum)
C) SELECT CUSTOMER-Name AND CUSTOMER-PhoneNum
D) SELECT (Name, PhoneNum)

2) Which of the following data types used in SQL would define a fixed-length text field of 7 characters?

A) Text(7)
B) Char(7)
C) VarDhar(7)
D) Fixed(7)

3) The number of entity classes involved in a relationship is known as the relationship's _____.

A) maximum cardinality
B) degree
C) ordinality
D) minimum cardinality

4) Which of the following is not a stage in the development of a database system?

A) Requirements analysis
B) Component design
C) Validation
D) Implementation

5) In many-to-many relationships in a relational database design, _____.

A) the keys of both tables are joined into a composite key
B) the key of the parent is placed as a foreign key into the child
C) the keys of both tables are placed in onty one of the tables
D) the key of the child is placed as a foreign key into the parent

AOU EGYPT

CamScanner